

Experiment : 1

Frequency Analysis and Probability in Data Science Using Python

Mapped CO(s) : CO1, CO2

Aim

To perform frequency analysis, calculate probability measures, and analyze correlation using the Titanic dataset in Python.

Objectives

- To create frequency tables for categorical variables.
- To compute absolute, relative, and cumulative frequencies.
- To calculate joint, marginal, and conditional probabilities.
- To construct contingency tables using Pandas.
- To analyze the correlation between numerical variables.
- To visualize relationships using heatmaps, scatter plots, and bar charts.

Prerequisites

- Basic knowledge of Python programming.
- Understanding of probability and statistics.
- Python 3.x installed.
- Jupyter Notebook or Google Colab.
- Required libraries: Pandas, NumPy, Matplotlib, and Seaborn.

Theory/Background

Frequency analysis is a statistical technique used to determine how often different values or categories occur in a dataset. A frequency table summarizes the distribution of categorical data by displaying the number of occurrences of each category. Relative frequency expresses these counts as percentages, while cumulative frequency represents the running total of frequencies.

A contingency table (or two-way table) displays the relationship between two categorical variables and is commonly used to calculate probability measures.

The probability measures used in this experiment are:

- **Joint Probability:** Probability that two events occur together.
- **Marginal Probability:** Probability of occurrence of a single event regardless of other variables.

- **Conditional Probability:** Probability of an event occurring given that another event has already occurred.

Correlation analysis measures the strength and direction of the relationship between two numerical variables. The Pearson correlation coefficient ranges from **-1 to +1**.

- **+1** indicates a perfect positive correlation.
- **-1** indicates a perfect negative correlation.
- **0** indicates no linear relationship.

Data visualization techniques such as heatmaps, scatter plots, and stacked bar charts help in understanding patterns and relationships within the dataset.

Algorithm/Methodology

Part I: Frequency Analysis

- Import the required libraries.
- Load the Titanic dataset.
- Generate a frequency table for the passenger class.
- Calculate absolute frequency.
- Calculate relative frequency (%).
- Calculate cumulative frequency.

Part II: Probability Analysis

- Create a contingency table between **Sex** and **Survived** using `pd.crosstab()`.
- Calculate joint probability.
- Calculate marginal probabilities.
- Calculate conditional probabilities.

Part III: Correlation Analysis

- Select the **Age** and **Fare** columns.
- Handle missing values.
- Compute Pearson correlation.
- Visualize the correlation using a heatmap.
- Plot a scatter plot for Age vs Fare.

Dataset Description

Dataset Name: Titanic Dataset

Source: Seaborn / Kaggle

Dataset URL:

<https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv>

The Titanic dataset contains demographic and travel information of passengers aboard the RMS Titanic. It is widely used for learning data analysis, statistics, and machine learning concepts.

Important Attributes

Attribute	Description
survived	Survival status (0 = No, 1 = Yes)
pclass	Passenger class (First, Second, Third)
sex	Gender of passenger
age	Age of passenger
fare	Ticket fare
embarked	Port of embarkation
class	Passenger travel class
who	Man, Woman, or Child
adult_male	Indicates whether passenger is an adult male
alone	Indicates whether passenger was traveling alone

Program/Implementation

The program should include:

- Importing Pandas, NumPy, Matplotlib, and Seaborn.
- Loading the Titanic dataset.
- Creating frequency tables.
- Computing relative and cumulative frequencies.
- Creating contingency tables using `pd.crosstab()`.
- Calculating joint, marginal, and conditional probabilities.
- Computing Pearson correlation.
- Generating heatmaps.
- Creating scatter plots.
- Creating stacked bar charts.

Test Cases

Test Case	Expected Result
Load Titanic dataset	Dataset loads successfully
Frequency table of Class	Absolute, relative and cumulative frequencies displayed
Contingency table	Sex vs Survived table generated
Joint probability	Correct probability calculated

Marginal probability	Correct probabilities displayed
Conditional probability	Conditional probabilities computed correctly
Pearson correlation	Correlation coefficient generated
Heatmap	Correlation matrix visualized
Scatter plot	Relationship between Age and Fare displayed
Stacked bar chart	Survival distribution by passenger class displayed

Experimental Results

Observation

- Frequency tables were successfully generated for passenger classes.
- Contingency tables clearly represented the relationship between gender and survival.
- Joint, marginal, and conditional probabilities were computed successfully.
- Missing values in the Age column were handled before correlation analysis.
- Pearson correlation between Age and Fare was calculated.
- Heatmaps and scatter plots effectively visualized the relationship between variables.
- The stacked bar chart showed differences in survival rates among passenger classes.

Result

Frequency analysis, probability calculations, and correlation analysis were successfully performed using the Titanic dataset. The visualizations enhanced the understanding of relationships between categorical and numerical variables.

Conclusion

The experiment demonstrated how frequency tables and probability concepts can be applied to real-world datasets using Python. Contingency tables helped compute joint, marginal, and conditional probabilities, while Pearson correlation measured the relationship between Age and Fare. Data visualization techniques such as heatmaps, scatter plots, and stacked bar charts made the analysis more meaningful and easier to interpret.

Exercises/Assignments

- Create a frequency table for the **Embarked** variable.
- Calculate the probability that a passenger survived.
- Find the Pearson correlation between **Age** and **Fare**.
- Create a stacked bar chart showing survival by passenger class.
- Interpret the obtained correlation coefficient.